DESIGN PROCESS DOCUMENTATION

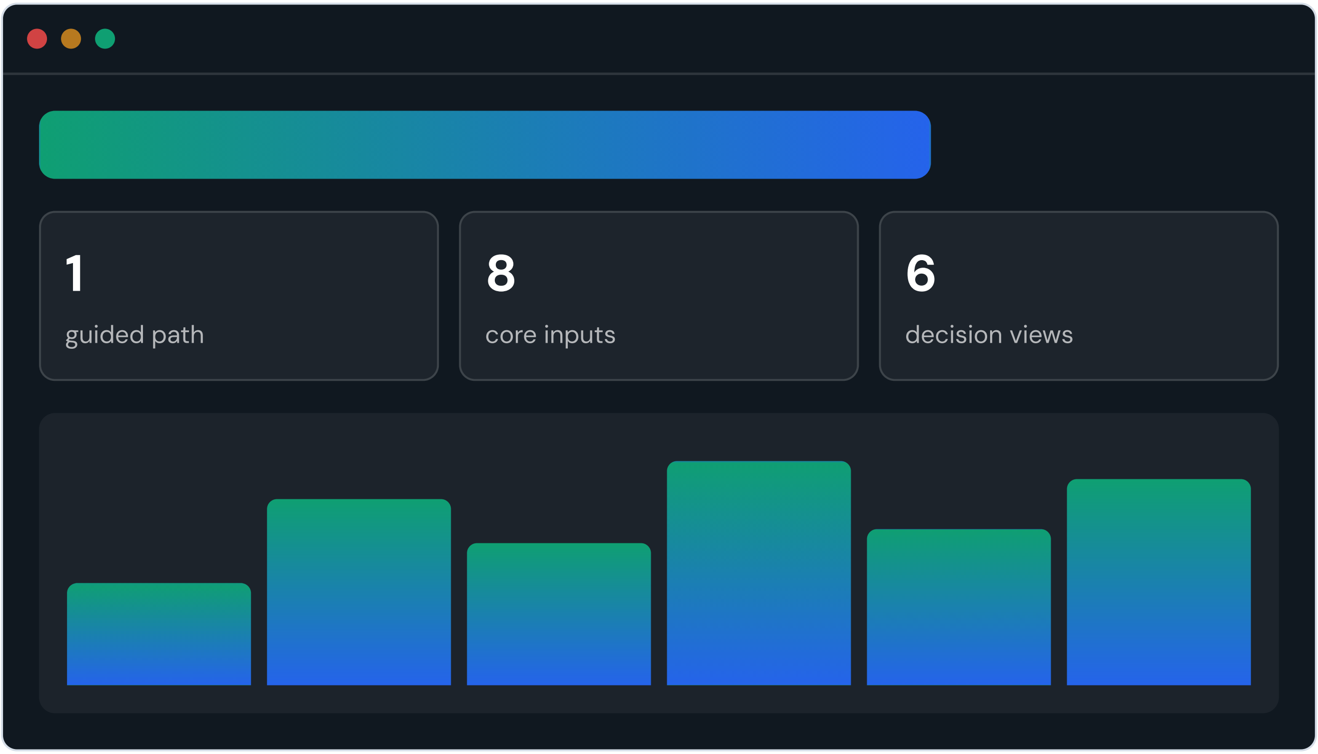
Making network optimization feel simple.

This document explains how the Vypaar Saathi prototype was designed: the user assumptions, research framing, heuristics, information architecture, visual language, interaction rules, and validation checklist.

Printable by design

Use the top-right button or browser print dialog and choose "Save as PDF". The document includes A4 print rules.

PDF ready



The product behaves like a guided decision tool first and an analytics platform second.

1. PRODUCT INTENT

Start with human confidence, then reveal analytical depth.

The prototype is for users who understand supplier pain but may not understand graph analytics or optimization terminology. The design goal is to make them feel oriented before asking for data.

Understand

What the tool does

The landing page states the offer and sends users to one guided start.

Enter

Only useful data

Sign-up collects just the fields required for meaningful downstream analysis.

See

Immediate outputs

KPIs, graph, charts, simulation, and optimizer all react to state changes.

Act

Choose the next move

The final output is a practical allocation recommendation, not a generic report.

2. USER ASSUMPTION

Designed for operators who need clarity, not complexity.

Business owner

Needs simple risk visibility

They want to know which supplier could hurt the business and what action reduces dependency.

Operations manager

Needs continuity planning

They care about lead time, recovery days, disruption pressure, and backup capacity.

Procurement lead

Needs explainable tradeoffs

They compare suppliers on cost, reliability, delivery, risk, compliance, and capacity.

3. JOURNEY SIMPLIFICATION

One user goal gets one obvious route.

The design removed duplicate setup paths and made the first-time journey linear. Advanced edits still exist, but they no longer compete with the guided start.

01

Landing

Clear product entry and showcase context.

02

Create

One guided sign-up, one question at a time.

03

Add info

Supplier and buyer data appears after basics.

04

Analyze

Dashboard, map, risk, and comparison update.

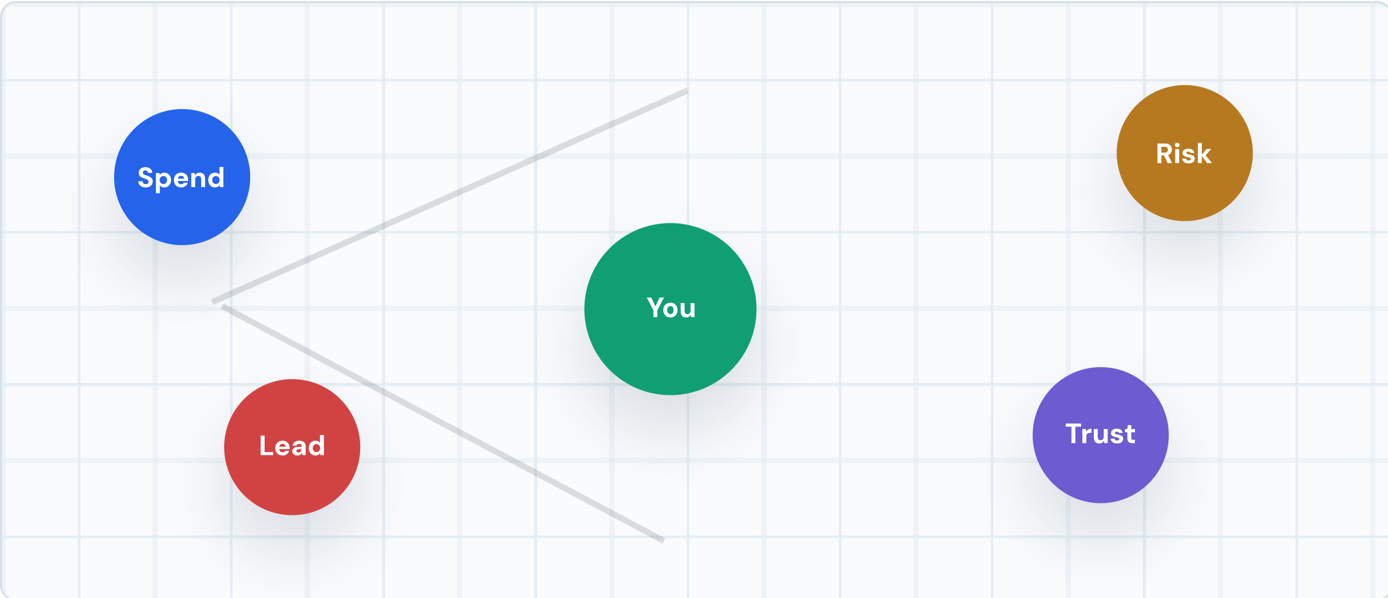
05

Optimize

User gets recommended allocation and reasons.

4. DATA RATIONALE

Every field has a job in the decision model.



Workspace

Company, role, critical category, annual spend, planning cycle, goal.

Partners

Name, type, risk level, reliability score, relationship weight.

Operations

Lead time, recovery days, disruption probability, capacity share.

Decision

Cost index, switching cost, compliance score, sharing preference.

5. HEURISTIC SCORECARD

Usability heuristics translated into product behavior.

Visibility of system status

Checklist, KPIs, progress dots, live badges.

Match with real world

Plain labels: Problems, Pick Best, Best Plan.

User control and freedom

Edit profile, add partners, reset, switch theme.

Recognition over recall

Suggested answers and visible setup progress.

Error prevention

Defaults, bounded numbers, simple selects.

Progressive disclosure

Basics first, deeper supplier data later.

Help and recovery

Reset path, editable data, no dead-end setup.

6. INFORMATION ARCHITECTURE

The navigation follows the user's thinking sequence.

Home

What should I do?

Start, progress, and simple result framing.

Map

Who am I connected to?

Spatial view of supplier and buyer relationships.

Pick Best

Which partner is better?

Weighted comparison and tradeoff testing.

Problems

What can go wrong?

Risk dashboard and Monte Carlo scenario outputs.

Best Plan

What should I change?

Recommended allocation, reasons, and constraints.

Add Info

What data is missing?

Profile, partner data, and sharing controls.

7. VISUAL DESIGN DIRECTION

Operational, calm, and decision-focused.

Plain language

The UI avoids model jargon until analytics need explanation.

Compact surfaces

Cards are used for decision units, not as decorative page containers.

Status color

Green, amber, red, and blue communicate safety, warning, risk, and action.

Light and dark themes

The prototype can be shown in different presentation environments.

Graph and charts

Visual analytics build credibility without making the user parse raw tables first.

Landing as entry

The landing page supports showcase needs, while the app remains the main experience.

8. INTERACTION RULES

A control is only useful if it changes the product.

Input to output

Completing sign-up updates profile, checklist, navigation state, and dashboard copy.

Partner add

Adding a supplier updates graph, KPIs, comparison choices, risk view, and optimizer.

Simulation run

Monte Carlo controls update stockout probability, expected loss, percentile loss, and service level.

Optimization

The best-plan view translates supplier data into allocation, score, reason, and constraint.

9. ACCESSIBILITY AND SIMPLICITY

Designed for low training burden.

Readable

Short labels

Navigation and actions use familiar business words.

Guided

One question

Sign-up asks one thing at a time with suggested answers.

Reversible

Edit paths

The user can reset, revise, and correct profile or partner data.

Visible

Status signals

Risk and progress are visible without requiring memory.

10. VALIDATION CHECKLIST

What must work before presenting the prototype.

New user can create workspace

Landing to guided sign-up.

Only one obvious setup route

No duplicate onboarding path on Home.

Partner data changes outputs

Graph, KPIs, risk, comparison, optimizer.

Monte Carlo run updates results

Scenario outputs visibly change.

Presentation and docs open

Deck, markdown, and this visual document are linked.

11. CURRENT LIMITS AND NEXT ITERATION

The design is ready for demo, then usability testing.

Current limitations

Local storage only, sample data, no auth, no CSV import, no production error handling, and simulation assumptions are illustrative.

Next design work

Add import flow, saved scenarios, executive summary export, persona-specific onboarding, graph accessibility, and usability testing with target users.

Vypaar Saathi prototype design documentation. Related files: `docs/design-process.md`, `docs/user-journey.md`, `docs/technical-report.md`, and `presentation.html`.